

Performance Testing

Date	4 July 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Performance Testing

This document evaluates the performance of the AI Powered Debt Relief & Financial Recovery Platform under normal operating conditions. It summarizes the testing approach, tools, environment, and observed results.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Browser Developer Tools, FastAPI, Uvicorn
Type of Testing	Load Testing, Functional Performance Testing
Target Module / API	User Login, Loan Management, Dashboard, AI Recommendation
Test Environment	Local System (Windows), React + FastAPI + SQLite
Test Date	15 March 2026

Step2: TestScenarios

S.No	Test Scenario	Virtual Users	Duration	Expected Outcome
1	User Login	1	30 sec	Login successful
2	Add Loan	1	45 sec	Loan saved
3	Dashboard Load	1	20 sec	Dashboard displayed
4	AI Recommendation	1	60 sec	Recommendation generated

Step 3: Performance Test Results

S.No	Metric	Target	Actual	Status	Remarks
1	Response Time (Avg)	<2 sec	1.2 sec	Pass	Good
2	Response Time (Max)	<5 sec	3.1 sec	Pass	Stable
3	Throughput	20 req/s	24 req/s	Pass	Acceptable
4	Error Rate	<1%	0%	Pass	No errors
5	CPU Utilization	<80%	42%	Pass	Normal
6	Memory Utilization	<80%	55%	Pass	Stable

Step 4: Observations & Analysis

KeyFindings:

- Application responded quickly under normal usage.
- Dashboard and loan operations completed successfully.

Bottlenecks Identified:

- Google Gemini API response depends on internet connectivity and API availability.

Optimization Steps Taken:

- Modular backend, optimized database queries, reusable React components, and FastAPIrouting.

Step 5: Screenshots / Evidence

Insert screenshotsof:

- Login Page
- Dashboard
- Loan Management
- AI Recommendation Results
- Browser Network/Performance Results